Specifications

01. Introduction

1.1 Context

The project aims to design an innovative connected system to facilitate the shopping experience for visually impaired or blind individuals. These individuals face daily challenges during their shopping, hindering their autonomy. The main goal of this solution is to increase their independence by integrating smart technologies into the process of grocery shopping.

1.2 Main concerns

Visually impaired or blind individuals often encounter obstacles during their shopping, such as product identification, navigation through aisles, and obtaining information about items. Our connected system, integrated with our mobile application, aims to overcome these challenges by providing intelligent support and facilitating access to information.

1.3 Project goals

The first objective is to develop a connected system that provides assistance with navigation and product search, allowing visually impaired or blind individuals to carry out their shopping independently. Integration with a dedicated mobile application aims to offer additional control and an enriched user experience. This project aligns with an inclusive approach, aiming to provide a practical and accessible solution.

1.4 Project scope

Where?

- The system operates in a supermarket (such as Carrefour).
- Large enclosed space, with several aisles that may change locations.
- Other customers and shopping carts are present.

For whom?

- Blind individuals.
- Visually impaired individuals.

2. Stakeholders

2.1 Blind users

Blind people will be the primary beneficiaries and end-users of the application. Their main interest is the gain of accessibility and usability within the application to navigate and shop independently in grocery stores. They have to provide feedbacks and suggest improvements to enhance user experience.

2.2 Developers

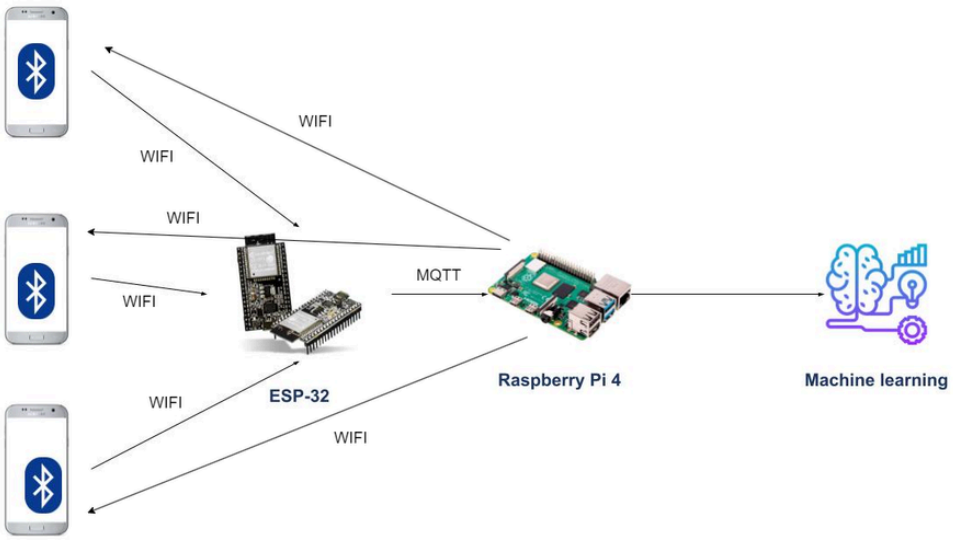
Developers will be responsible of the design, the development and the maintenance of the application. Their main interest is to meet user requirements, ensure technical feasibility and deliver a high-quality, accessible product.

2.3 Grocery stores owners

Grocery stores owners will be responsible to host the application within their stores. Their main interest is to enhance shopping experience for all customers, including those with visual impairments and ensure smooth integration of the technology into store operations. They have to collaborate with developers to implement the application in-store, address any logistical challenges, and provide feedback on its effectiveness.

3. System architecture

The solution proposed is the one below.



In a grocery store, blind people will try to localize themselves using our application. Our application will connect to an ESP-32 using the WIFI protocol. To allow the ESP-32 to communicate with the Raspberry Pi 4, messages will be sent using the MQTT protocol. This protocol is a set of rules that defines how IoT devices can publish and subscribe to data over the Internet. All relevant user's data will be collected via the ESP-32 and sent in destination of the Raspberry Pi 4. With the user's data, the Raspberry will be responsible of the calculs and the mapping of the user's in our map. It will then sent the position of the user through his phone.

4. Functional requirements

4.1 User stories

Id:	As:	I'd like to:	What for:
1	A blind person	Be guided through the story aisles	Find the products on my shopping list
2		Enter my shopping list	Be directed to these products
3		Know my location within the store	Be able to be guided in real-time
4		Be able to verbally communicate with the application	Use the application

5		Have a vocal feedback	Facilitate the readability of displayed information
6		Update the status of my shopping list after picking up an item	Be guided to the next product/aisle
7		Save my shopping list	Reuse it later
8		Add items from my list to a basket	View the products I'm going to purchase, as well as the total amount
9		Search for a product sold by the store	View the products I'm going to purchase, as well as the total amount
10		Sorting by: unit price, price per kg	Check aisle availability
11		Filter by: brand, nutri-score	Refine my selection criteria
12	Shop staff	Fill in the shop plan	Locate shop customers
13		Fill in a JSON file representing products in my shop (database)	Allow the user to choose a product knowing it's in the shop

4.2 Additional requirements

Additional requirements may be incorporated into the user stories to ensure the comprehensive functionality of the application.

Requirements
Users must have at least a mobile phone with Bluetooth (BLE).
Users must have access to a voice assistant.
Grocery owners must setup all IOT devices.
Grocery owners must fill in the plan of their shop into the application.
Grocery owners must give informations concerning their products into the store.

5. Non functional requirements

5.1 Performance Requirements:

The application should have a response time of less than 5 seconds for any user interaction to ensure a smooth and efficient user experience.

Location tracking accuracy should be within 3 meters to provide precise navigation guidance within the store environment.

5.2 Accessibility Requirements:

The application interface must be compatible with standard screen readers and assistive technologies commonly used by blind and visually impaired individuals.

All user interface elements, including buttons, labels, and navigation controls, should be designed with sufficient contrast and size to facilitate usability for users with low vision.

5.3 Security Requirements:

User data, including personal information and shopping history, must be encrypted both in transit and at rest to ensure confidentiality and integrity.

6. User interface Design

6.1 Description of user interface design principles

The user interface (UI) design will adhere to principles of simplicity, intuitiveness, and clarity to ensure an easy and efficient shopping experience for blind users.

Auditory feedback will guide users through the interface, providing clear cues and instructions.

6.2 Mockups

Detailed mockups and wireframes will be created to visualize the layout and functionality of the application.

These visual representations will depict key UI elements, such as navigation menus, product search interfaces, and shopping list management screens.

6.3 Guidelines for ensuring accessibility and usability for blind users

The UI will prioritize accessibility features to accommodate blind and visually impaired users effectively.

Text-based navigation options and voice commands will be provided as alternatives to visual cues, allowing users to interact with the application using speech or touch gestures.

All UI elements will be labeled with descriptive text to facilitate screen reader compatibility, ensuring that blind users can navigate the interface seamlessly.

High contrast color schemes and large, legible fonts will be employed to enhance readability and visibility for users with low vision.

User feedback mechanisms will be implemented to gather input from blind users and iteratively improve the accessibility and usability of the interface.

7. System Integration and Testing

7.1 Integration plan for Hardware and Software components

The integration plan will outline the steps and timeline for combining hardware (ESP32, Raspberry Pi) and software (mobile application) components into a cohesive system.

It will detail the installation, configuration, and setup process for each component, ensuring compatibility and seamless communication between them.

Regular integration checkpoints will be established to monitor progress and address any compatibility issues or dependencies.

7.2 Testing

The testing methodology will be divided in four main concerns:

- Unit Testing: Individual software components, including modules and functions, will undergo unit testing to verify their correctness and functionality in isolation.
- Integration Testing: Once individual components are tested, integration testing will be conducted to validate the interaction and communication between hardware and software elements.
- End-to-End Testing: Comprehensive end-to-end testing will assess the entire system's behavior and functionality under real-world conditions, including location tracking, navigation, and user interaction.
- Regression Testing: Regression tests will be performed to ensure that system updates or modifications do not introduce new defects or regressions in existing functionality.

7.3 Acceptance criteria for system testing

The system should accurately track the user's location within the store environment, with deviations within an acceptable margin of error (±0,5 meters).

Navigation instructions provided by the application should guide users to their desired destinations accurately and efficiently, minimizing deviations and unnecessary detours.

The user interface should respond promptly to user inputs and interactions, with minimal latency or delays.

The application should be compatible with standard screen readers and assistive technologies commonly used by blind and visually impaired individuals, ensuring accessibility and usability.

The system should gracefully handle errors and exceptions, providing informative error messages and guiding users through recovery steps when necessary.

8. Deployment and Maintenance

8.1 Deployment Plan:

The deployment strategy will begin with a controlled pilot phase in selected grocery stores, gradually expanding to wider adoption based on user feedback and performance evaluations.

Comprehensive training materials and workshops will be developed to educate users, including blind individuals and store personnel, on utilizing the application effectively for navigation and product identification.

Strategic marketing campaigns will be launched to generate awareness and encourage adoption of the ShopForMe application, leveraging digital platforms, community engagement, and partnerships with relevant organizations.

8.2 Maintenance Plan

Regular updates will be deployed to enhance the application's capabilities.

A dedicated support team will be available to assist users with any issues, providing timely resolution of inquiries and troubleshooting assistance.

Ongoing monitoring and user feedback will inform iterative enhancements, ensuring that the application remains intuitive, efficient, and responsive to users' needs.

8.3 Scalability and future enhancements

The architecture will be designed to accommodate future growth and scalability, with provisions for handling increased user demand and expanding functionality.

In the future, AI-driven product recognition capabilities will be integrated into the application, allowing users to scan products using their mobile device's camera and receive real-time audio feedback, thereby augmenting their ability to independently navigate and shop in stores.

Conclusion

In summary, the ShopForMe project aims to revolutionize the grocery shopping experience for visually impaired and blind individuals by leveraging smart technologies and innovative design principles. By providing intelligent support for navigation, product identification, and information access, the application empowers users to shop independently and with confidence.

Through collaborative efforts with stakeholders, including blind users, developers, and grocery store owners, the project aligns with an inclusive approach to address the unique challenges faced by visually impaired individuals in their daily lives.

With a focus on continuous improvement and scalability, future enhancements will introduce AI-driven product recognition capabilities, further enhancing the application's functionality and usability.

In conclusion, the ShopForMe project embodies the principles of accessibility, innovation, and empowerment, aiming to make a meaningful difference in the lives of visually impaired individuals and foster a more inclusive society. The journey continues with ongoing development, feedback integration, and the pursuit of excellence in delivering a transformative shopping experience for all.